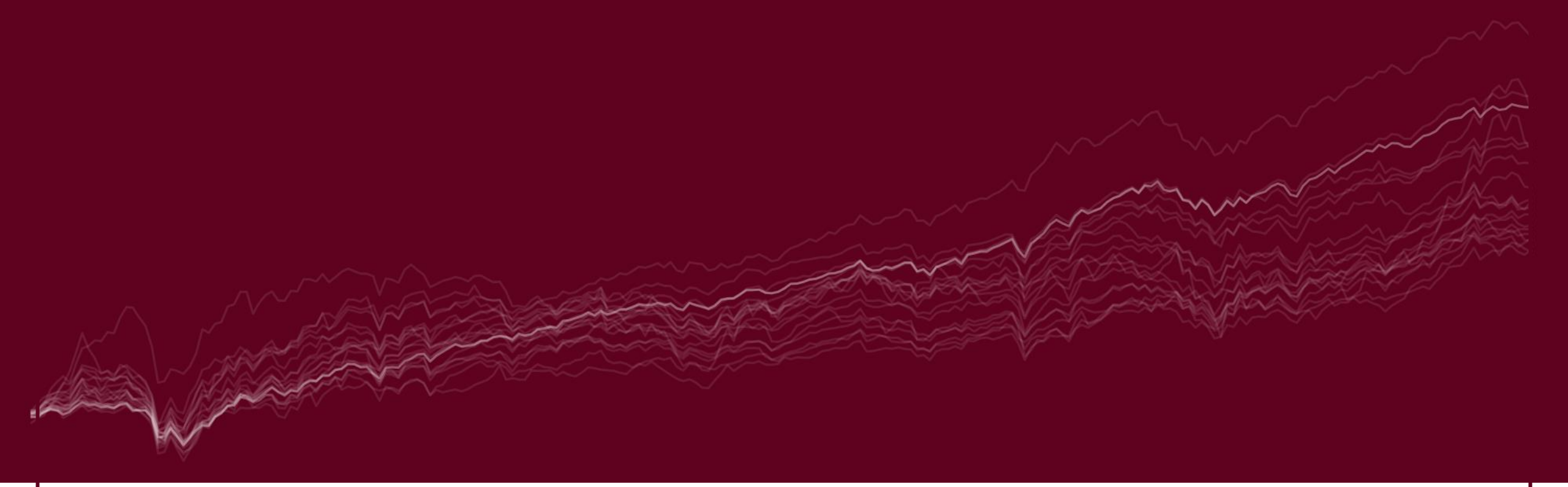


Regime-Diversified Mean Reversion

DOES THE STRATEGY TRAVEL? C-A ON 19 OTHER EQUITY MARKETS

Public Markets Research · November 2026

These slides are for educational purposes only. All results are backtests on historical data from 2007 to 2026 and are in-sample in time. They do not constitute financial, legal or investment advice, and no guarantee is made regarding their accuracy or completeness. Past performance does not predict future returns.

Overview of Today's Session

AGENDA

1. Recap: the C-A strategy on the S&P 500
2. Why the two mechanisms should work elsewhere
3. Research design and four registered hypotheses
4. Results: headline, dip overlay, rebalancing, costs
5. Verdict and honest caveats
6. Q&A

19

equity markets besides the S&P 500

3

regions: US segments, developed ex-US, emerging

20 yrs

of daily data, 2007-02 to 2026-11

The Strategy: C-A

RECAP OF THE FROZEN RULES

Book 1 · Index Dip Overlay

50 % OF CAPITAL

- Hold the index at exposure 1.0
- **Dip:** RSI(2) < 10 and close > SMA200 → exposure 2.0 until close > SMA5
- **Overbought:** RSI(2) > 90 and close < SMA200 → exposure 0.5 until close < SMA5

50/50

Book 2 · Rebalancing Premium

50 % OF CAPITAL

- Index, long Treasuries (TLT) and gold (GLD) in equal thirds
- Reset to 1/3 each at every month end
- Weights drift with prices in between

On SPY, 2007–2026: Sharpe 0.78 vs 0.53 for buy & hold. Holdout 2023-11 to 2026-11: 1.42 vs 1.09, paired p = 0.165.

Decision at the close of day t, held on day t+1 · 1 bp per side · Sharpe on excess returns over the 3-month T-bill

Why It Should Travel

ECONOMIC THEORY

Book 1: liquidity provision

Forced or hurried selling pushes the index below fair value. Whoever buys the dip is paid for providing liquidity when the reversal comes. This should happen in any market where investors sometimes sell in a hurry.

Book 2: diversification return

Rebalancing sells relative winners and buys relative losers. The gain is mechanical and grows with the volatility of the assets:

$$DR \approx \tfrac{1}{2}\left(\Sigma\, w_i \sigma_i^{\,2} - \sigma_p^{\,2}\right)$$

Registered prediction: the dip overlay should be weaker in emerging markets, because their indices show more short-horizon momentum and USD-listed ETFs mix currency moves into their dips.

Lehmann (1990); Nagel (2012); Booth & Fama (1992); Erb & Harvey (2006)

Research Design

ONE CHANGE, EVERYTHING ELSE FROZEN

- **Only the equity asset changes,** in book 1 and in book 2
- TLT, GLD, every threshold, costs and cash accounting stay as they are
- US-dollar investor, US-listed ETFs, 4,970 trading days
- Hypotheses committed to git before the first number was computed
- Pooled tests use the effective number of independent markets, n_eff

US SEGMENTS

QQQ Nasdaq-100

IWM Russell 2000

MDY S&P MidCap 400

DIA Dow Jones 30

OEF S&P 100

DEVELOPED EX-US

EFA MSCI EAFE

EZU Eurozone

EWG Germany

EWU United Kingdom

EWJ Japan

EWA Australia

EWC Canada

EMERGING

EEM MSCI Emerging

EWZ Brazil

FXI China large caps

EWY South Korea

EWT Taiwan

EWW Mexico

EZA South Africa

💡 **Same window everywhere:** all 20 series share the same 4,970 trading days without gaps. India is excluded because no ETF covers the full window.

Four Registered Hypotheses

SUCCESS CONDITIONS FIXED IN ADVANCE

	Hypothesis	Economic cause	Success condition
M1	C-A beats buy & hold of the local market	Treasuries and gold diversify any equity market	≥ 15 of 19 markets and pooled $p < 0.10$
M2	The dip overlay (book 1) beats buy & hold	Liquidity-provision premium exists everywhere	≥ 13 of 19 and pooled $p < 0.10$, in both 2007–15 and 2016–26
M3	Monthly rebalancing beats never rebalancing	Diversification return is mechanical	≥ 13 of 19 and rank correlation with volatility > 0
M4	The result survives realistic costs	Turnover is only ~10× per year	At 5 bp per side, at most 2 markets fewer than at 1 bp

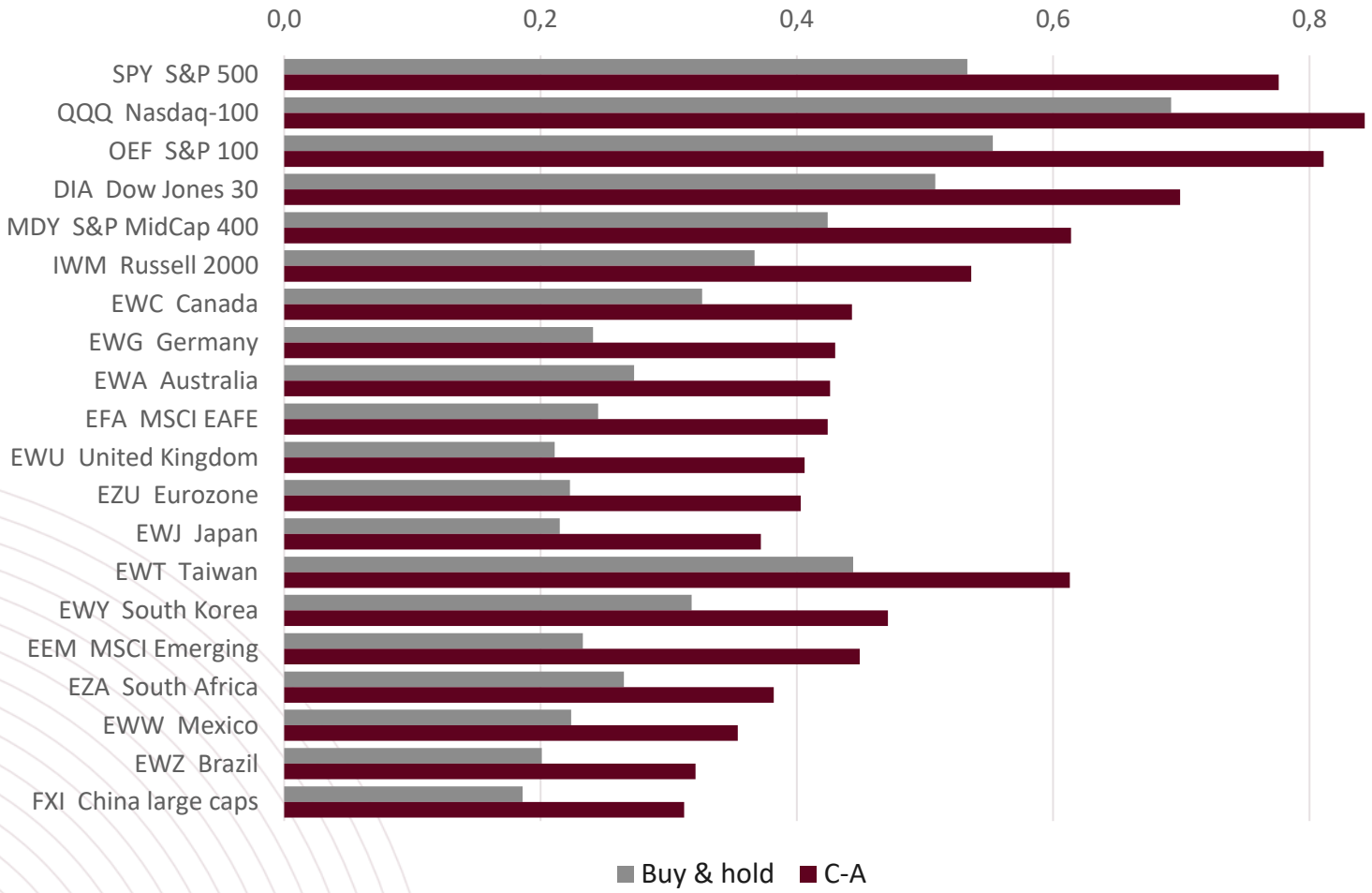
🔑 **"C-A generalises" = M1 and M2.** M1 alone would only say that bonds and gold diversify equities, which is not news.

Results

19 MARKETS · 2007-02 TO 2026-11

C-A vs Buy & Hold

SHARPE RATIO 2007–2026, EXCESS OVER T-BILL, USD



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markets: C-A has the higher Sharpe

p = 0.011

pooled across markets

n_eff = 1.3

differences are 73 % correlated: one common cause

The Same Pattern in Every Region

AVERAGES BY GROUP

Group	C-A Sharpe	Buy & hold	Book 1 – B&H	C-A max DD	B&H max DD	Break-even
Reference (SPY)	**0.78**	0.53	+0.15	−32 %	−55 %	38 bp
US segments (5)	**0.70**	0.51	+0.09	−34 %	−55 %	32 bp
Developed ex-US (7)	**0.42**	0.25	+0.09	−41 %	−62 %	33 bp
Emerging (7)	**0.42**	0.27	+0.10	−50 %	−69 %	37 bp

~20 pp

smaller maximum drawdown than buy & hold, in every group

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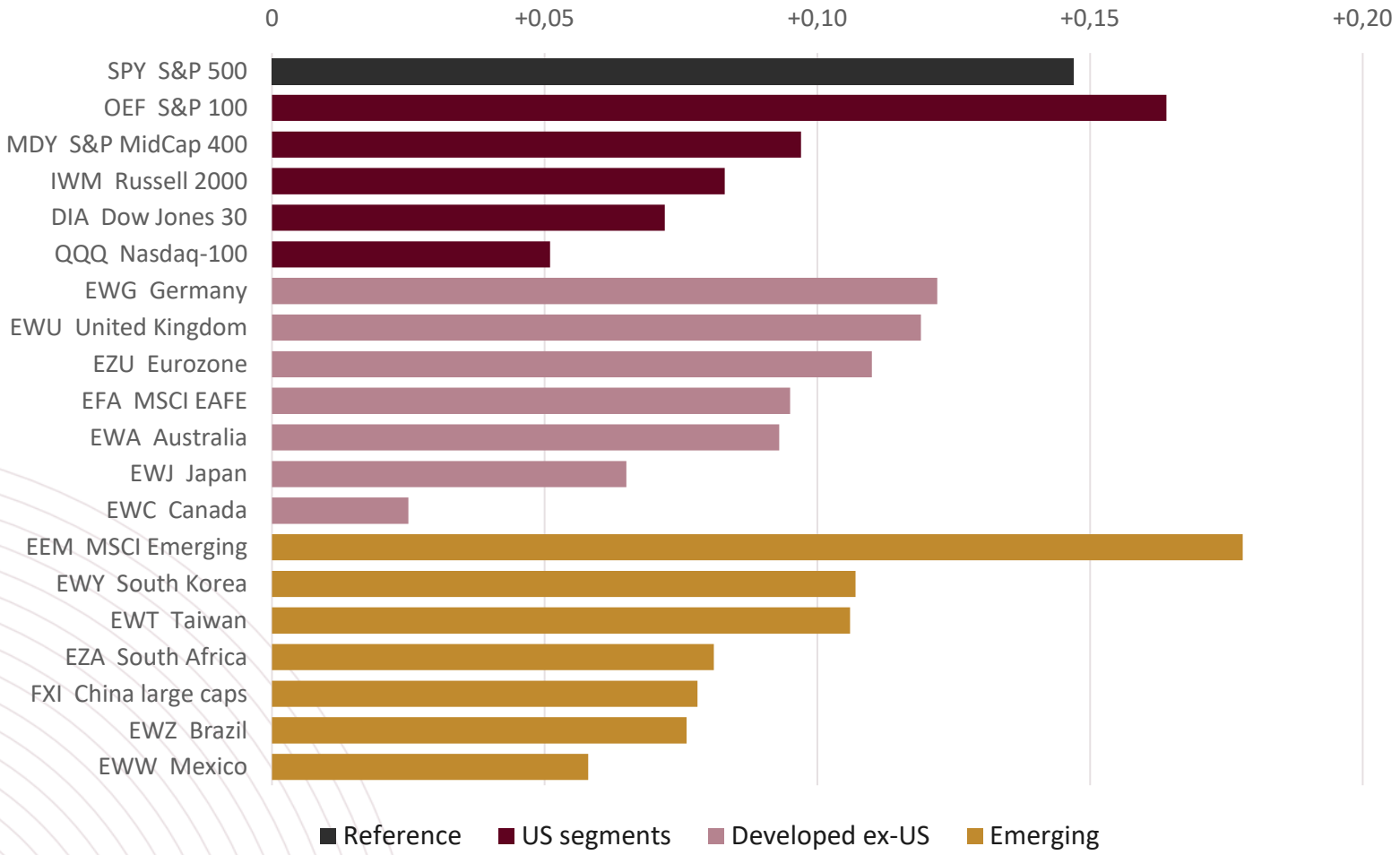
markets with a higher CAGR too (exception: QQQ, 15.3 % vs 15.6 %)

0.42 vs 0.78

C-A abroad stays far below C-A on SPY: the market's own Sharpe matters most

The Dip Overlay

BOOK 1 MINUS BUY & HOLD, SHARPE 2007–2026



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markets: the overlay adds Sharpe

2 of 19

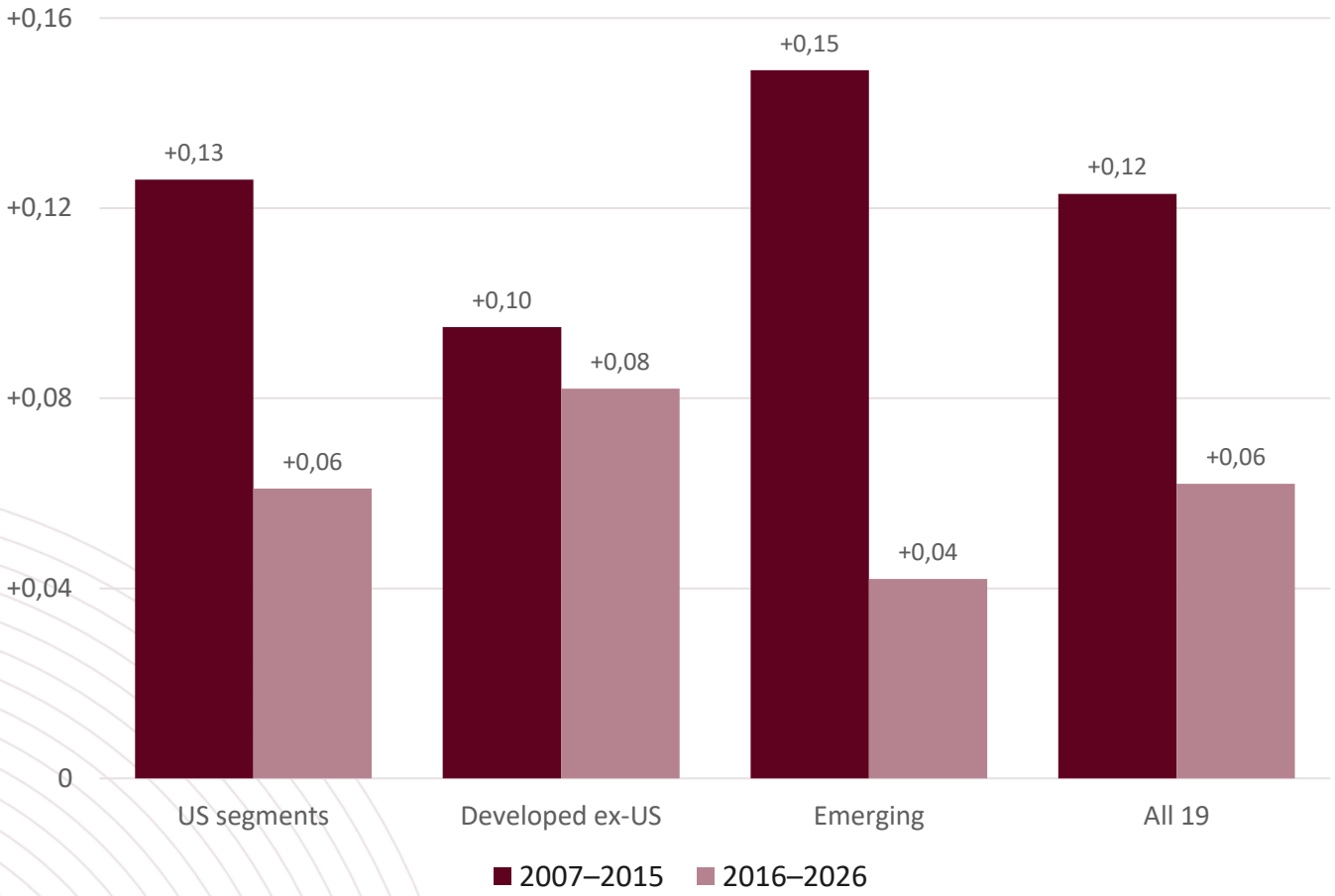
significant on their own (OEF, EEM)

p = 0.024

pooled, n_eff = 2.6 (differences only 35 % correlated)

The Overlay Faded After 2016

AVERAGE SHARPE GAIN OF BOOK 1 OVER BUY & HOLD



Period	Positive	Pooled p
2007–2015	18 / 19	0.043
2016–2026	17 / 19	**0.27**

M2 fails: it needs pooled $p < 0.10$ in both halves. After 2016 the gain halves and is no longer distinguishable from noise.

Same decay as on SPY: index reversal was stronger before 2016.

Our Emerging-Market Prediction Was Wrong

PREDICTION VS OUTCOME

REGISTERED

Weaker in EM

More short-horizon momentum in EM indices and currency noise in USD-listed ETFs should work against reversal.

OBSERVED

EEM strongest of all

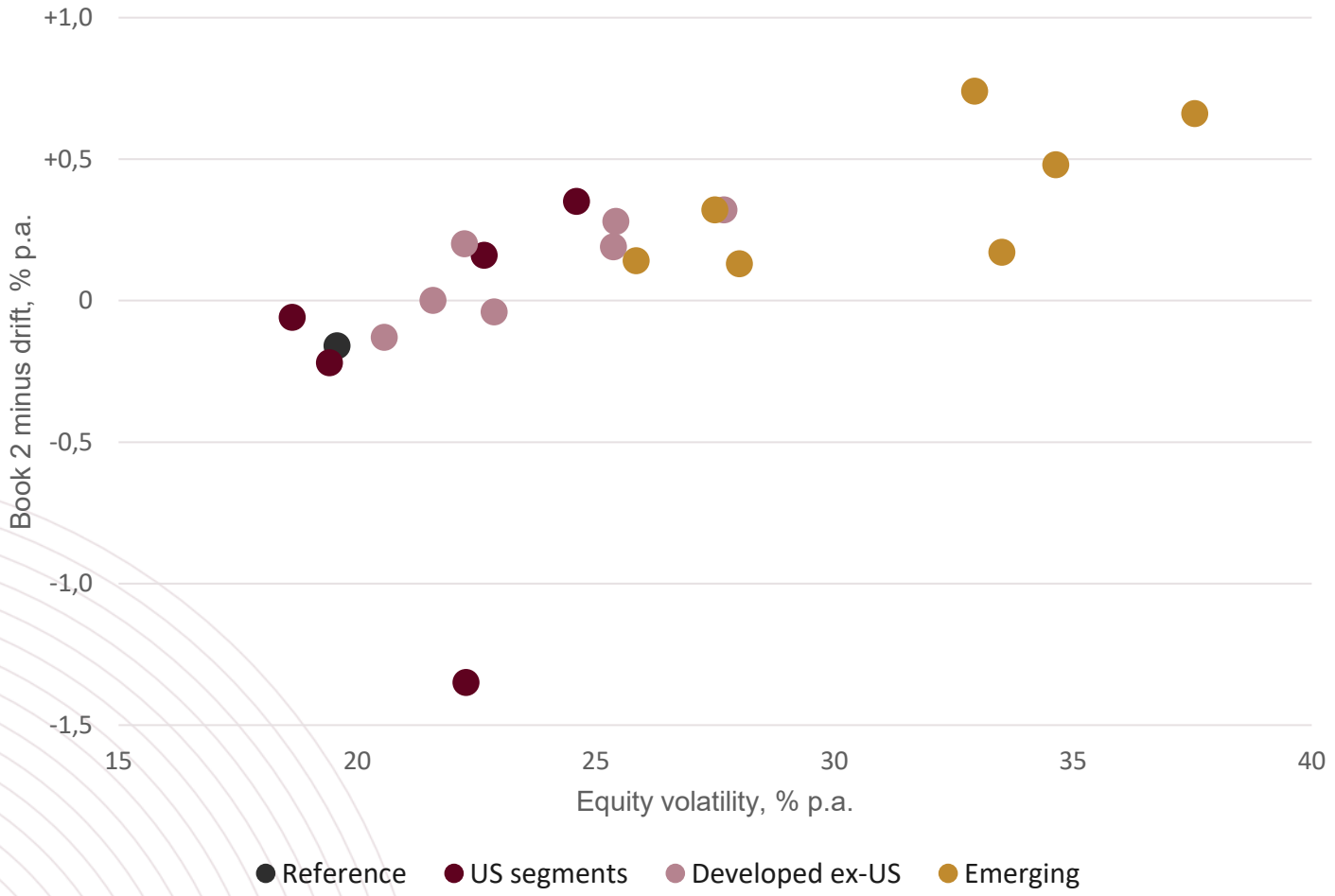
+0.18 Sharpe from the overlay, p = 0.003. The emerging group has the strongest pooled evidence (p = 0.02).

A possible explanation (untested)

- A US-listed EM ETF trades while its local markets are closed, so US-hours selling can push it below the value of its holdings
- That gap closes the next day: an ETF-microstructure effect, not the index reversal C-A is built on. Testing it needs NAV or local-index data

Rebalancing Scales with Volatility

BOOK 2 MINUS NEVER-REBALANCED, RETURN % P.A.



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markets: higher Sharpe with monthly rebalancing

$\varrho = +0.74$

rank correlation with equity volatility, as theory predicts

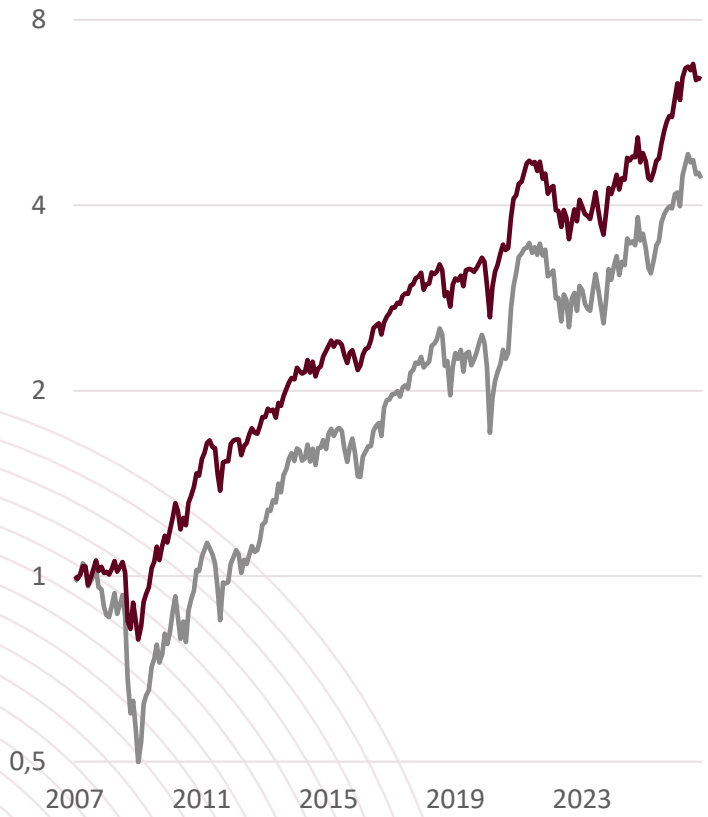
- **Korea, Brazil, South Africa:** +0.5 to +0.7 % p.a.
- **QQQ (bottom):** −1.3 % p.a., the Nasdaq rally rewarded letting it run
- **2016–2026:** only 8 of 19, letting gold run paid

Growth of 1 USD

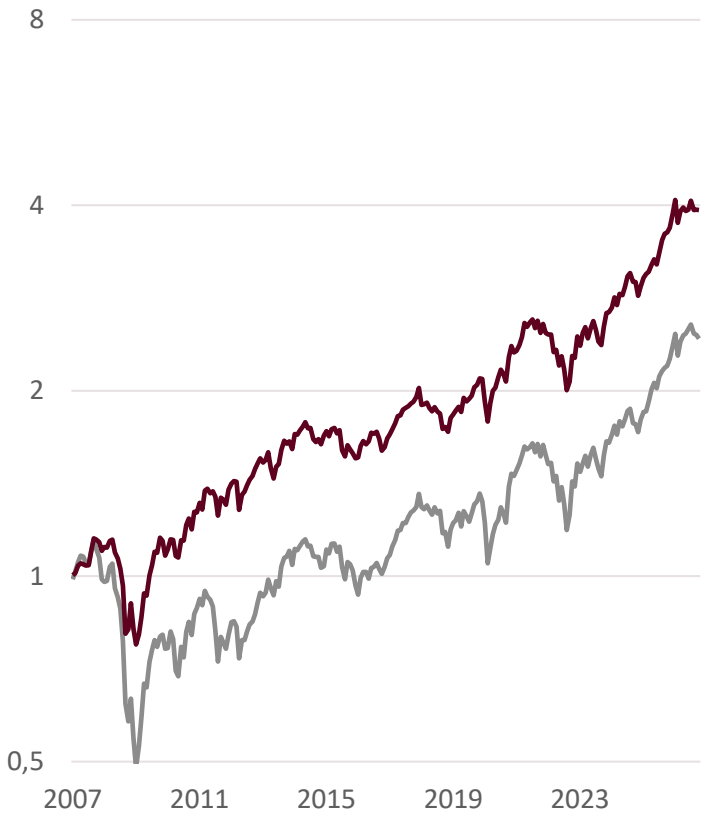
C-A VS BUY & HOLD, LOG SCALE

C-A Buy & hold

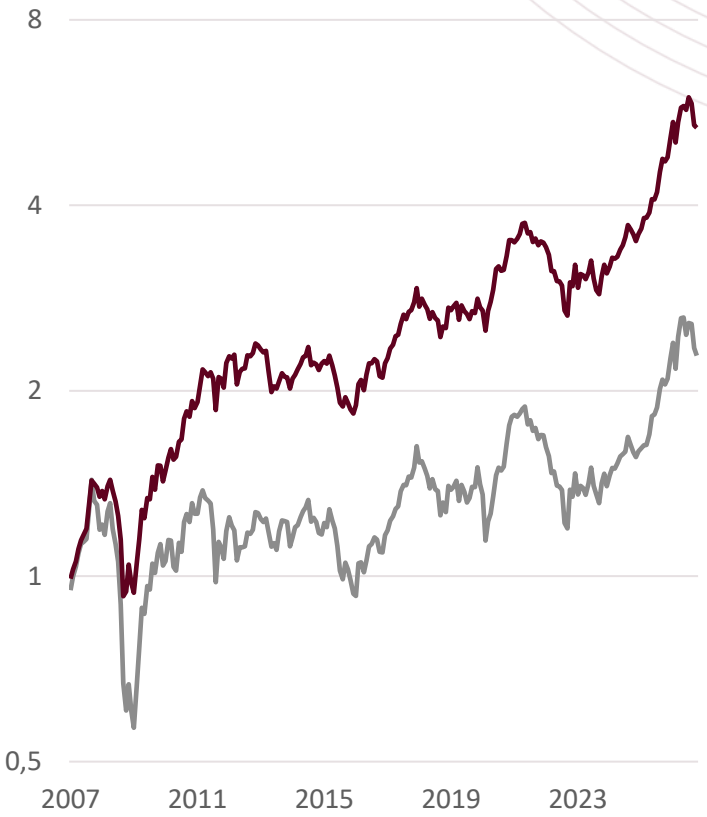
Russell 2000 (IWM)



MSCI EAFE (EFA)



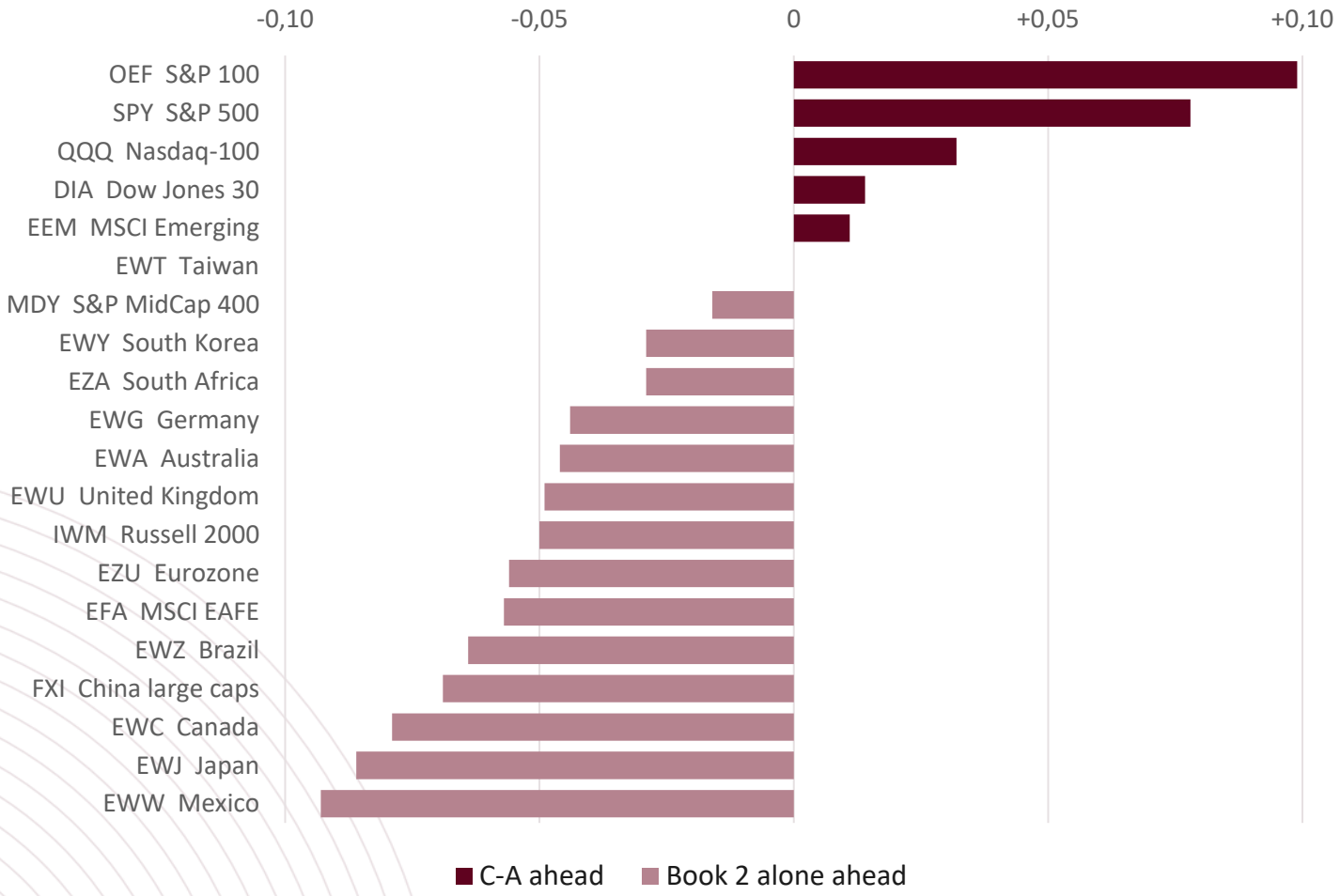
MSCI Emerging (EEM)



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But: Book 2 Alone Was Often Better

C-A MINUS BOOK 2 ALONE, SHARPE · NOT REGISTERED



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markets where book 2 alone has the higher Sharpe

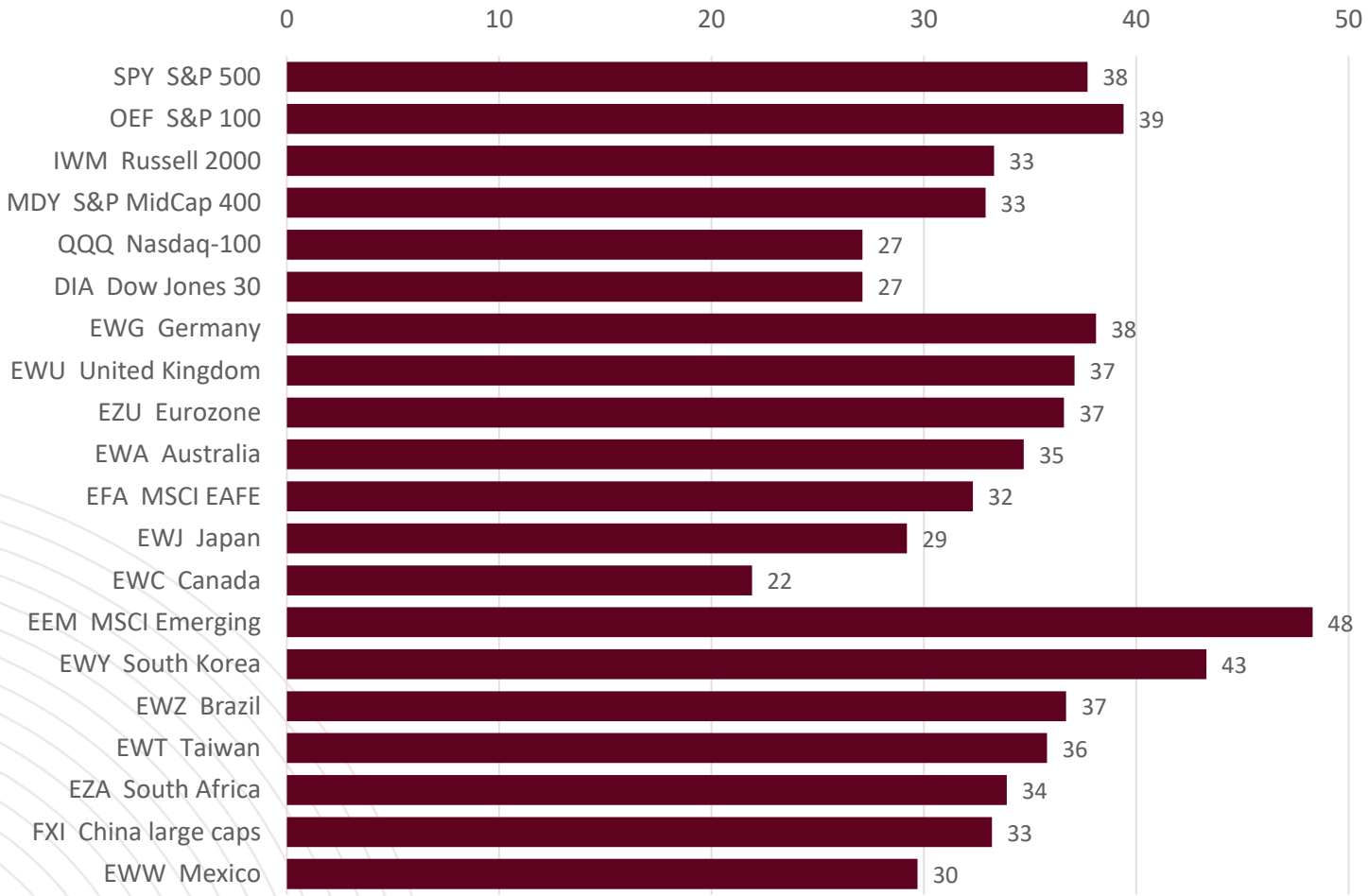
p = 0.77

pooled: none of the differences is significant

💡 **No contradiction:** the overlay improves an equity position everywhere, but where equities had a low Sharpe, less equity was better.

Costs Are Not the Problem

BREAK-EVEN COST PER SIDE VS BUY & HOLD, BASIS POINTS



22–48 bp

per side before the edge over buy & hold disappears

1–5 bp

realistic cost of trading these ETFs

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markets still ahead at 5 bp per side (M4 passes)

Verdict

THE FOUR REGISTERED TESTS

M1	**C-A beats buy & hold**	19 / 19, pooled p = 0.011	PASSED
M2	**Dip overlay beats buy & hold, both halves**	2007–15: p = 0.043 · 2016–26: p = 0.27	FAILED
M3	**Rebalancing beats never rebalancing**	19 / 19, ρ(volatility) = +0.74	PASSED
M4	**Survives 5 bp per side**	19 / 19, break-even 22–48 bp	PASSED

“C-A generalises” (M1 and M2): not met. The rebalancing book travels; the dip overlay travels in sign, but not in strength.

Honest Caveats

WHAT THIS STUDY DOES NOT SHOW

In-sample in time

Every day from 2007 to 2026 has been looked at. The markets are out-of-sample only in the cross-section.

Correlated witnesses

The 19 headline wins count as 1.3 independent markets. They share one cause: the Treasury and gold half.

A prior partial run

Six of the 19 markets overlap an earlier eight-country check. The registration missed it; an erratum is in the repository.

Survivorship and USD lens

Only ETFs that exist today. Returns include currency moves. Yahoo revises adjusted prices slightly over time.

Takeaways

WHAT FOLLOWS FROM THE STUDY

1

Rebalancing travels

Its premium shows up in every market and grows with volatility. This half of C-A does not depend on the US.

2

The dip overlay travels in sign, not in strength

Positive in all 19 markets, but the evidence sits before 2016. It is no reason to trade C-A abroad instead of on SPY.

3

C-A on SPY stays the strategy

Nothing here changed its rules. Next: the forward test, and a newly registered study of the EEM gain with NAV data.

Code, data and the registration: github.com/RobinHumberg/regime-diversified-mean-reversion

Q&A